

Problem 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^2 \sin x$. Compute the derivative:

$$\begin{aligned} f'(x) &= 2x \sin x + x^2 \cos x \\ &= x(2 \sin x + x \cos x). \end{aligned}$$

Problem 2

Show that

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

Problem 3

Simplify $\sqrt{x^2 + 2x + 1}$ for $x \geq -1$:

$$\begin{aligned} \sqrt{x^2 + 2x + 1} &= \sqrt{(x+1)^2} \\ &= |x+1| \\ &= x+1 \quad \text{since } x \geq -1. \end{aligned}$$